

Xinyuan Yan, Ph.D.

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EDUCATION

Postdoctoral Researcher, Neurosurgery & Neuroscience April 2025 – Present
Baylor College of Medicine *Advisors: Benjamin Y. Hayden & Sameer A. Sheth*

Postdoctoral Researcher, Psychiatry & Neuroscience March 2022 – March 2025
University of Minnesota *Advisors: Alexander Herman & David Darrow*

Ph.D., Cognitive Neuroscience September 2016 – January 2022
IDG/McGovern Institute for Brain Research, State Key Laboratory of Cognitive Neuroscience and Learning
Beijing Normal University

B.Sc., Psychology September 2012 – June 2016
Northeast Normal University

FELLOWSHIPS & AWARDS

Travel Award, Computational and Systems Neuroscience (Cosyne), Montreal	2023
Trainee Professional Development Award, Society for Neuroscience	2022
MnDrive Neuromodulation Research Fellowship, University of Minnesota Medical School	2022
Fellowship, Computational and Cognitive Neuroscience (CCN) Summer School, CSHL Asia	2021
Fellowship, Kavli Summer Institute in Cognitive Neuroscience, UC Santa Barbara	2019
Distinguished Graduate Research Fellowship, State Key Laboratory of Cognitive Neuroscience and Learning	2018
National Graduate Research Fellowship, China	2018

PUBLICATIONS

† indicates co-last author. * indicates co-first author.

Concepts, language, single-neuron recordings, and hippocampus

1. **Yan X**, Chavez AG, Franch M, ..., Hayden BY†, Sheth SA† (2026). Shared neural geometries for bilingual semantic representations in human hippocampal neurons. *Cell*.
2. **Yan X**, Li JA, Franch M, ... Hayden BY†, Sheth SA† (2026). Polysemanticity in human hippocampal neurons. *bioRxiv*.
3. Chericoni A, Diao C, **Yan X**, ... Sheth SA†, Hayden BY† (2026). Neural geometry in the human hippocampus enables generalization across spatial position and gaze. *arXiv. Neuron (in press)*.
4. Xiong HD*, **Yan X***, Li JA, Xue J, Mattar MG, Wilson RC (2026). Thinking Under Uncertainty: Evidence Use and Information-Seeking in Language Models. *arXiv preprint*.
5. Chavez AG, **Yan X**, Franch M, ... Hayden BY (2026). Hippocampus serves as a repository for spoken and heard word meanings during conversations. *bioRxiv*.
6. Ismail T, Chavez AG, **Yan X**, ... Hayden BY (2026). Estimation of neuronal tuning for word meaning from passively recorded naturalistic speech. *bioRxiv*.

Concepts about uncertainty and implications for affective disorders

Skill set: computational modeling, intracranial EEG

7. **Yan X**, Peterson CM, Schmidt LM, Koenig S, Nixdorf DR, Herman A, Darrow D (2026). Impaired adaptive learning in chronic pain contributes to apathy. *eLife*, 2026-01.
8. **Yan X**, Ebitz RB, Grissom N, Darrow DP, Herman AB (2025). Distinct computational mechanisms of uncertainty processing explain opposing exploratory behaviors in anxiety and apathy. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
9. Jin T, Zhang S, **Yan X**, Vilares I (2026). Social Influence in Donation Behavior: The Effect of Mean, Variance, and Psychopathy. *eLife*, 2026-08.
10. **Yan X**, Ebitz BR, Darrow DP, Herman AB (2025). Nonlinear modulation of human exploration by distinct sources of uncertainty. *bioRxiv*, 2025-04.
11. **Yan X**, König SD, Ebitz RB, Hayden BY, Darrow DP, Herman AB (2025). Dynamic prefrontal coupling coordinates adaptive decision-making. *Research Square*, rs-3.
12. Knep E, **Yan X**, Chen CS, Jacob S, Darrow DP, Ebitz RB, . . . Herman AB (2025). Social aloofness is associated with non-social explore–exploit decisions. *Communications Psychology*, 3(1), 106.
13. Cao D*, Cui J*, **Yan X***, Zhang X, Dang Y, Zhao H, . . . Zhang Y (2025). Dynamic interactions in human amygdala subregions support hierarchical processing of fearful faces. *bioRxiv*, 2025-11.

Social-Concepts & Social Cognition

Skill set: computational modeling, fMRI

14. **Yan X**, Yong X, Huang W, Ma Y (2018). Placebo treatment facilitates social trust and approach behavior. *PNAS*, 115(22), 5732–5737.
15. Liu Y, Li S#, Lin W#, Li W#, **Yan X**, Wang X, Pan X, Rutledge RB, Ma Y (2019). Amygdala activity tracks deviations from social preferences. *Nature Neuroscience*, 22(4), 633–641. [# co-second author]
16. Wang D, **Yan X**, Li M, Ma Y (2017). Neural substrates underlying oxytocin effect: A quantitative meta-analysis of pharmaco-imaging studies. *Social Cognitive and Affective Neuroscience*, 12(10), 1565–1573.

Self-Concepts and implications for major depressive disorder

Skill set: fMRI

17. **Yan X** (2024). Neural dynamics of self-referential processing and the insight for decoding self-concepts. *Journal of Neuroscience*, 44(30). [Commentary]
18. Feng C†, **Yan X†**, Huang W, Han S, Ma Y (2018). Neural representations of the multidimensional self in the cortical midline structures. *NeuroImage*, 183, 291–299. [† co-first author]

Cortisol and the stress model

Skill set: resting-state fMRI

19. Feng Z, Zhang H, Zhou T, Yu X, Zhang Y, **Yan X*** (2023). Dynamic functional connectivity changes associated with psychiatric traits and cognitive deficits in Cushing’s disease. *Translational Psychiatry*, 13(1), 308. [* corresponding author]

20. Shang G, Zhou T, **Yan X**, . . . Zhang Y (2025). Multiscale analysis reveals hippocampal subfield vulnerabilities to chronic cortisol overexposure: Evidence from Cushing's disease. ***Biological Psychiatry: Cognitive Neuroscience and Neuroimaging***.
21. Liu B, Mao Z, **Yan X**, Yang H, Xu J, Feng Z, Zhang Y, Yu X (2024). Structural network topologies are associated with deep brain stimulation outcomes in Meige syndrome. ***Neurotherapeutics***, 21(4), e00367.

EDITORIAL SERVICE

- **Guest Associate Editor**, *Frontiers in Human Neuroscience* — Research Topic: *Methods in Cognitive Neuroscience: 2023* (Oct 2023 – Dec 2023)

AD-HOC REVIEWER

PNAS · Nature Communications · Communications Biology · Translational Psychiatry · Brain and Behavior · Psychoradiology (Oxford Academic) · Self and Identity · Frontiers in Psychology · Frontiers in Psychiatry · Frontiers in Human Neuroscience · Frontiers in Endocrinology · Frontiers in Robotics and AI · BMC Neuroscience.

PROFESSIONAL MEMBERSHIPS

- Society for Neuroscience
- Society of Biological Psychiatry (Regular Member)

CONFERENCE PRESENTATIONS

Oral Presentations, Invited Talks

- **Yan X.**, *Stable meaning, flexible codes: how the human hippocampus builds and reuses semantic geometry*. NYU Langone Health, August 5th, 2026, New York, USA.
- **Yan X.**, *Shared neural geometries for bilingual semantic representations in human hippocampal neurons*. NYU Psychology, August 6th, 2026, New York, USA.
- **Yan X.**, Koneig S, Ebitz B, Hayden, B, Darrow D, Herman A. *Dynamic prefrontal coupling coordinates adaptive decision-making*, Society for Neuroscience, November 15-19, 2025, San Diego, USA.
- **Yan, X.**, König, S., Ebitz, B., Hayden, B., Darrow, D., & Herman, A. (2024). *Deciphering neural mechanisms of explore–exploit decision-making in the prefrontal cortex*. Computational Psychiatry Conference, Minneapolis, USA.
- **Yan, X.**, Ebitz, B., . . . Darrow, D., & Herman, A. (2022). *The behavioral profile of the exploration–exploitation dilemma and its implications for anhedonia and anxiety*. Computational Psychiatry, New York City, USA.

Poster Presentations

- **Yan X**, *Shared neural geometries for bilingual semantic and grammatical representations in human hippocampal neurons*. 9th annual conference on cognitive computational neuroscience. 2026, August 6th, New York, USA.
- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2023). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Computational and Systems Neuroscience (Cosyne), Montreal, Canada.

- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2022). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Society for Neuroscience Annual Meeting, San Diego, USA.

ADDITIONAL TRAINING

- Computational and Cognitive Neuroscience (CCN) Summer School, *Cold Spring Harbor Laboratory*, 2021
- Online Summer School for Computational Neuroscience, *Neuromatch Academy*, 2020
- Kavli Summer Institute in Cognitive Neuroscience, *UC Santa Barbara*, 2019
- Coursera coursework (2020): Computer Simulations · Social Network Analysis · Network Dynamics of Social Behavior · Julia Scientific Programming · Facial Expression Recognition with Keras · Introduction to HTML5

TEACHING

Mentoring junior graduate students in core skills for scientific research:

- Representational similarity analysis
- Resting-state fMRI analysis
- Diffusion decision model & value-based DDM
- Reinforcement learning & Markov decision processes
- Attractor networks & decision-making
- Neural networks-basic concepts
- Building a personal academic website (HTML5 + CSS3)

GRADUATE STUDENTS/TRAINEES SUPERVISED:

- Ana G. Chavez
- Assia Chericoni
- Taha Ismail
- Will Baltazar